

The chart integral as a genuine d -dimensional integral:

$$\int_{(0,b]^d} \prod u_i^{h_i} f(c \prod u_i^{k_i}) du = \text{iterChartGen } d \ c$$

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Abstract

The recursive `iterChartGen` of unit 50 integrates one coordinate at a time; the paper writes the standard integral over the box $(0, b]^d \subseteq \mathbb{R}^d$. This unit defines the box integral as a Bochner integral on $\text{Fin } d \rightarrow \mathbb{R}$ against the product of the restricted Lebesgue measures and proves it equal to the recursion, so that the multiplicity theorems of units 50 and 52 hold verbatim in paper form (`boxIntegral_isEquivalent`, `boxIntegral_mixed_isEquivalent`). Zero sorry/axiom.

1 Setting

Splitting off the last coordinate of $\text{Fin}(d+1) \rightarrow \mathbb{R}$ is the measurable equivalence `piFinSuccAbove` at `Fin.last d`, which is measure preserving from the product of the $d+1$ restricted measures to the product of one restricted measure with the product of the other d (`measurePreserving_piFinSuccAbove`). Fubini on this product reduces the $(d+1)$ -fold integral to $\int_0^b u^{h_d} (d\text{-fold integral at scale } cu^{k_d}) du$, which is the recursion.

2 The things formalised

```
noncomputable def boxMeasure (b : ℝ) (d : ℕ) : Measure (Fin d → ℝ) :=
  Measure.pi fun _ => (volume : Measure ℝ).restrict (Ioc 0 b)
noncomputable def boxIntegral (a b c : ℝ) (k h : ℝ → ℝ) (d : ℕ) : ℝ :=
  u : Fin d → ℝ, ( i, u i ^ h i) * quadKernel a (c * i, u i ^ k i) (boxMeasure b d)
theorem boxIntegral_eq_iterChartGen (a b : ℝ) (k h : ℝ → ℝ) (d : ℕ) (c : ℝ) :
  boxIntegral a b c k h d = iterChartGen a b k h d c
theorem boxIntegral_isEquivalent ... :
  (fun n => boxIntegral a b (Real.sqrt n) k h (d + 2)) ~[atTop]
  fun n => ( i Finset.range (d + 2), (1 : ℝ) / k i)
    * (weightedMass a (p - 1) / (2 ^ (d + 1) * ((d + 1).factorial : ℝ)))
    * (n ^ (-(p / 2)) * Real.log n ^ (d + 1))
theorem boxIntegral_mixed_isEquivalent ...
```

3 Proof strategy

Induction on d generalising the scale c . For $d = 0$ the product measure over the empty index is a Dirac measure (`Measure.pi_of_empty`) and the integral is the kernel at c . For the step, the integrand is rewritten as $g \circ e$ with $g(t, v) = (\prod_j v_j^{h_j}) t^{h_d} f(c \prod_j v_j^{k_j} t^{k_d})$ using

`Fin.prod_univ_castSucc` and the two definitional facts $(eu).1 = u(\text{last})$, $(eu).2\ j = u(\text{castSucc } j)$ (`Fin.succAbove_last_apply`); then `MeasurePreserving.integral_comp'` and `integral_prod`. Integrability of g on the product of restricted measures is obtained by identifying that measure with Lebesgue measure restricted to the box (`Measure.prod_restrict`, `Measure.restrict_pi_pi`) and using continuity on the compact $[0, b] \times [0, b]^d$. The inner integral is the induction hypothesis at scale cu^{k_d} after pulling out u^{h_d} .

4 Roadblocks and resolutions

- `simp` would not unfold the structure-literal coercion of `piFinSuccAbove`; the two component identities are `rfl` up to `succAbove_last_apply`, so state them as `haves` and `simp only` with them under the product binder.
- `fun_prop` does not know `quadKernel`; assemble continuity by hand from `continuous_finsetProd`, `continuous_apply`, and `quadKernel_continuous`.
- `Measure.volume_pi` is not the name on this `pi`; since `volume` on a `pi` type is definitionally `Measure.pi`, a change to the `Measure.pi` form suffices before `restrict_pi_pi`.

5 What was Mathlib, what was new

Mathlib: `measurePreserving_piFinSuccAbove`, `MeasurePreserving.integral_comp'`, `integral_prod`, `Measure.pi_of_empty`, `integral_dirac`, `Measure.restrict_pi_pi`, `Fin.prod_univ_castSucc`, `isCompact_univ_pi`. New: the box integral and its identification with the recursion.

6 Lessons

Keeping the recursion as the working definition and proving the paper-form identity once at the end was the right order: all analysis was done on one-dimensional objects, and the d -dimensional measure theory is confined to a single induction.

7 Outcome

The multiplicity theorems for the chart standard integral with arbitrary (k, h) are now stated on $(0, b]^d \subseteq \mathbb{R}^d$ exactly as in the paper, in the all-equal and one-noncritical regimes.